

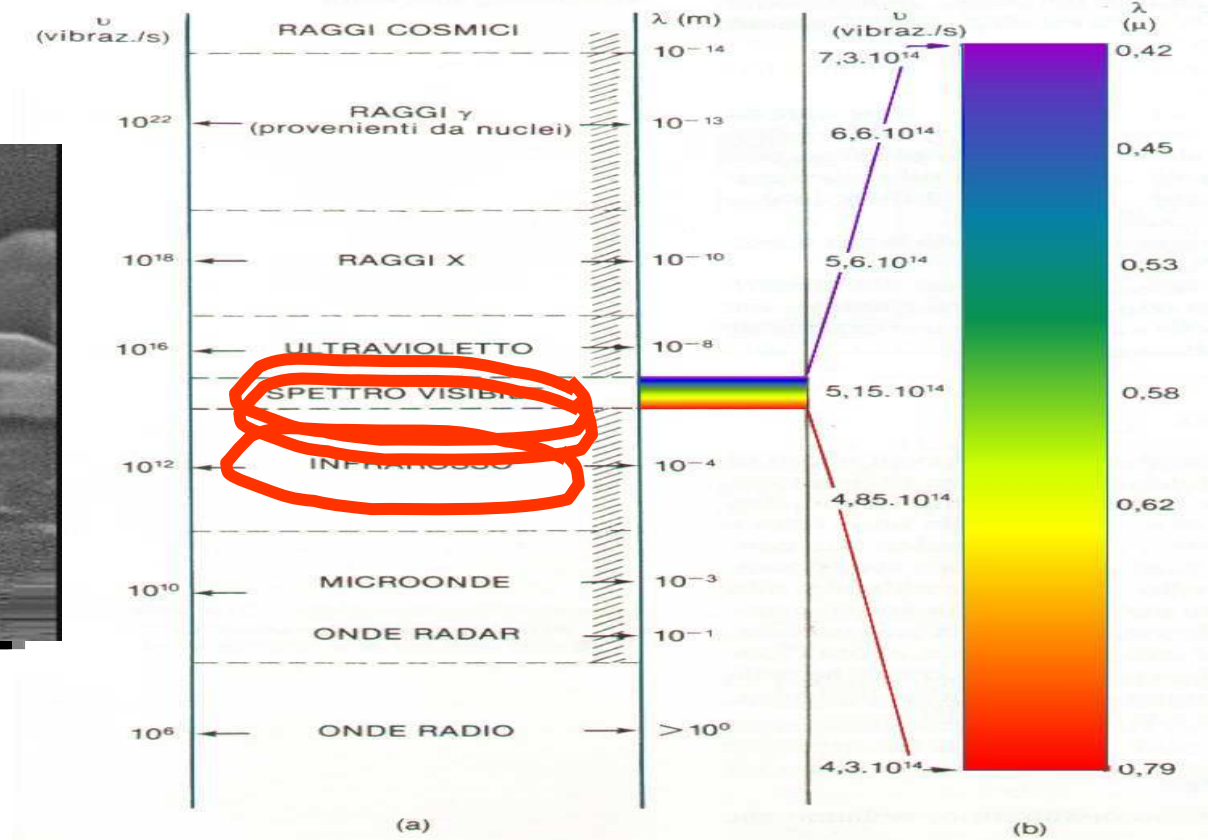


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# Video & Acquisition Systems

- Light
- Acquisition pipeline
- CCD vs CMOS
- Color extraction techniques

# Light Spectrum



# How do we sense light?

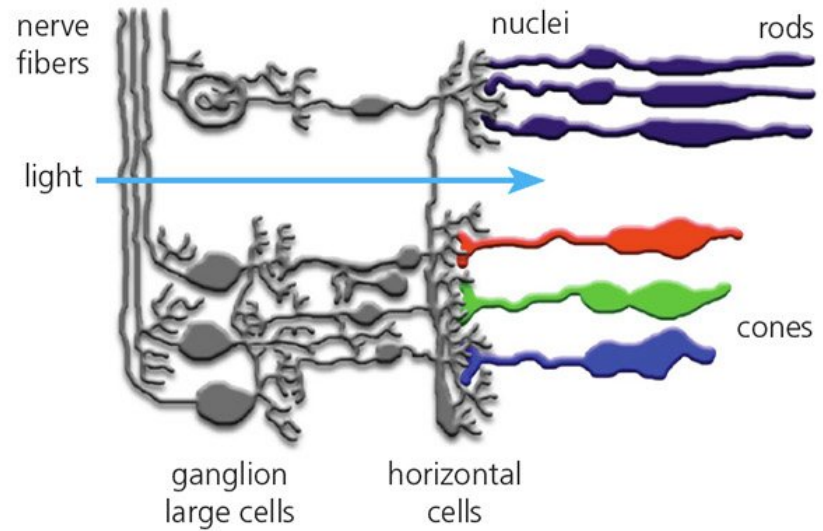
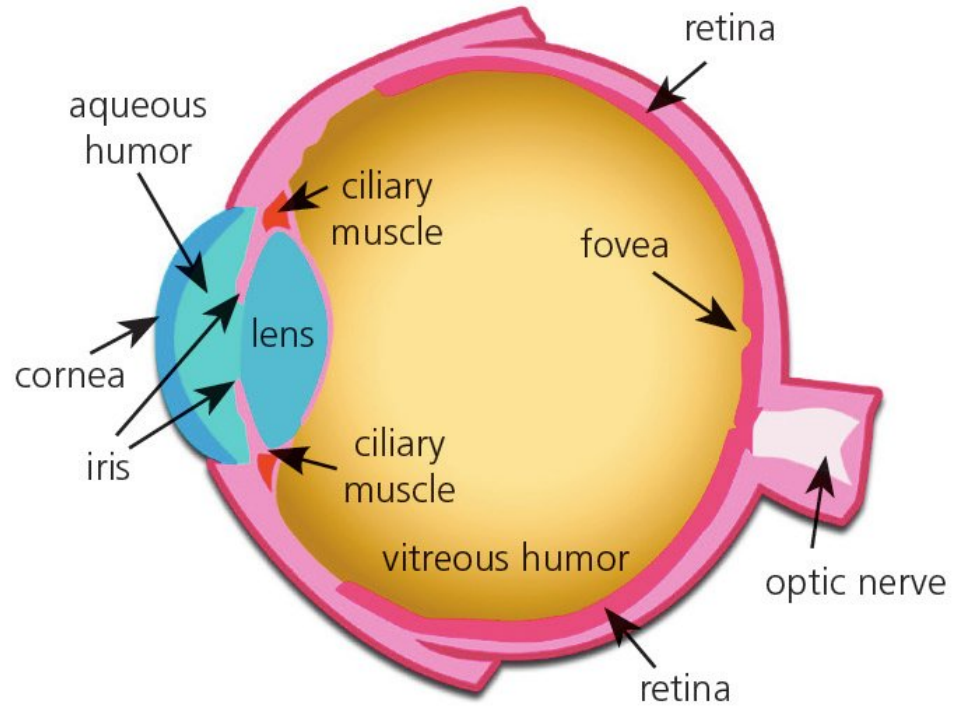




# How do we sense light?

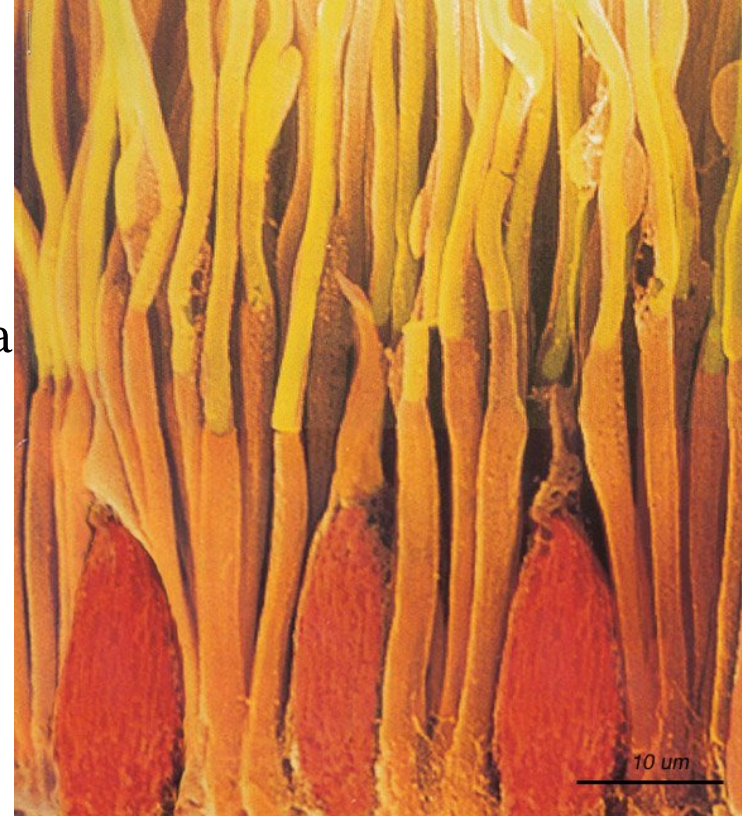


# Human Vision

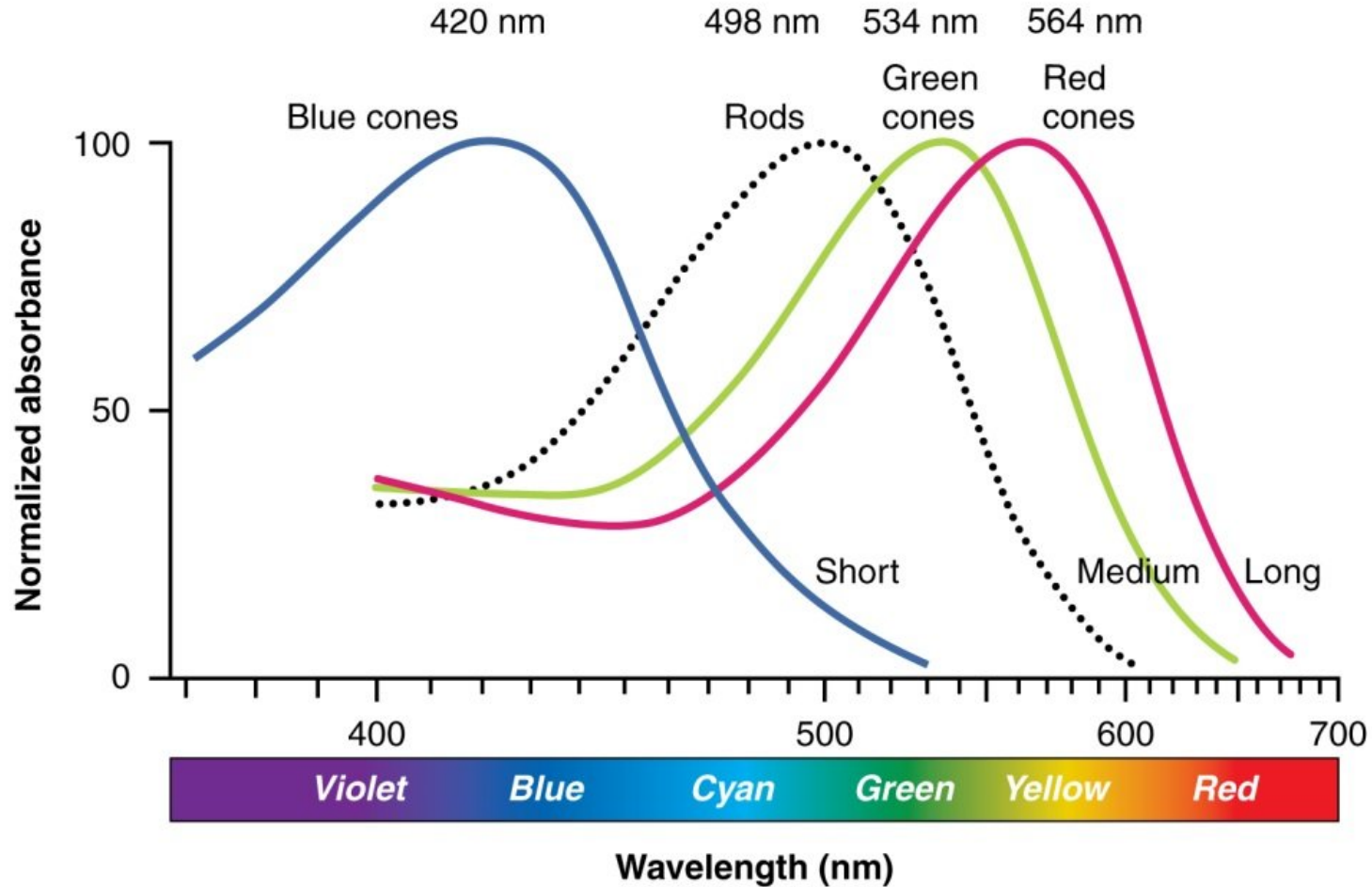




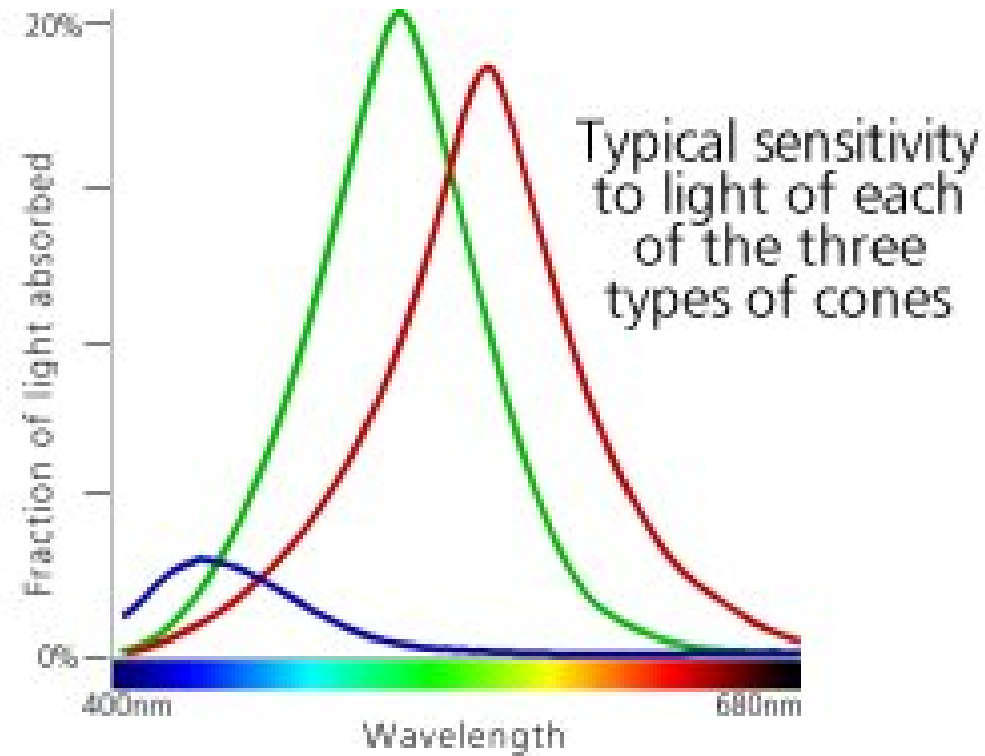
- “Two” sensors
- Rods (~100M)
  - More sensitive
  - Mostly distributed on peripheral part of retina
- Cones (~4-7M)
  - “Band pass filters”
  - Different types (typically 3)
  - Mostly around fovea

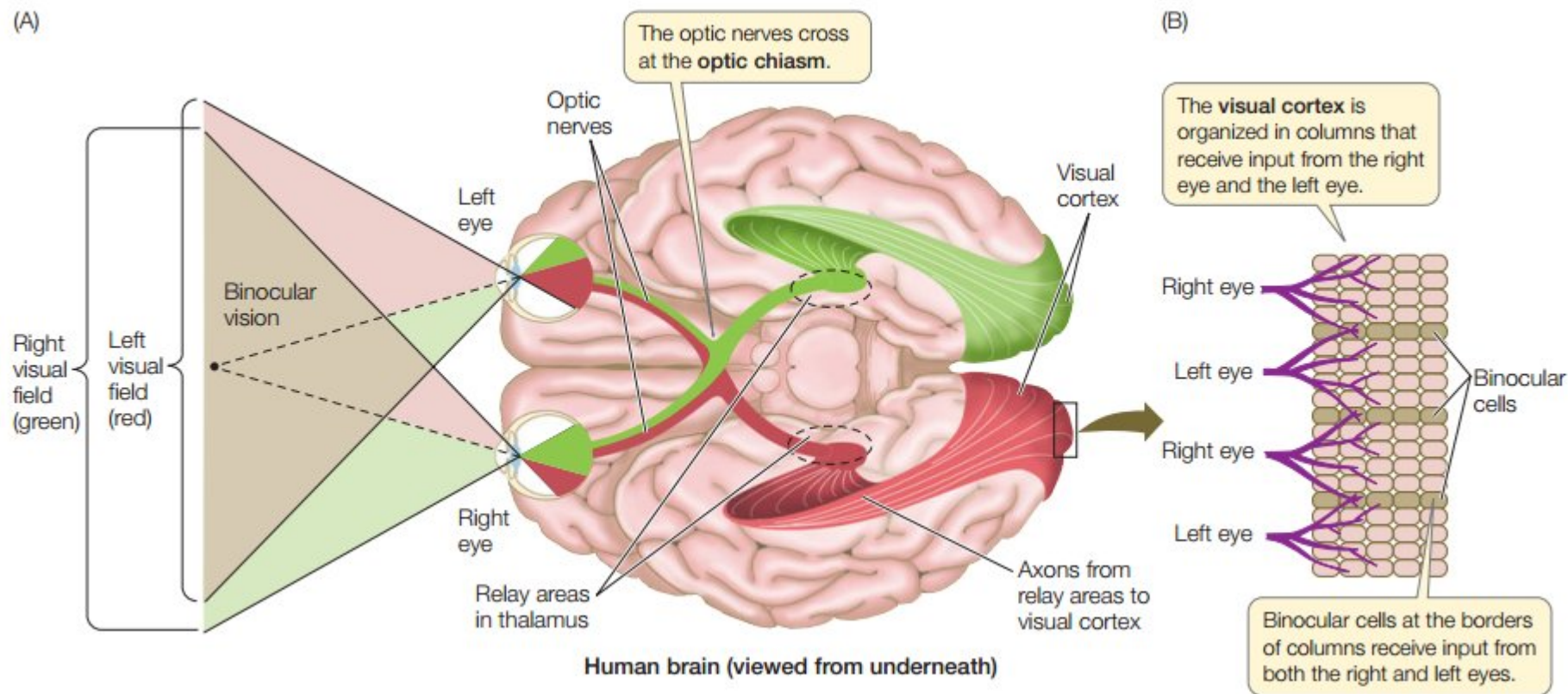


*Fig1b. Scanning electron micrograph of the rods and cones of the primate retina. Image adapted from one by Ralph C. Eagle/Photo Researchers, Inc.*

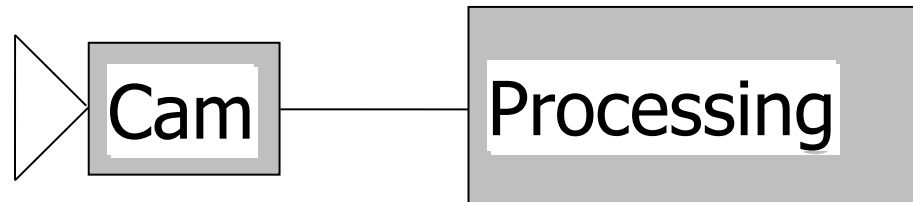






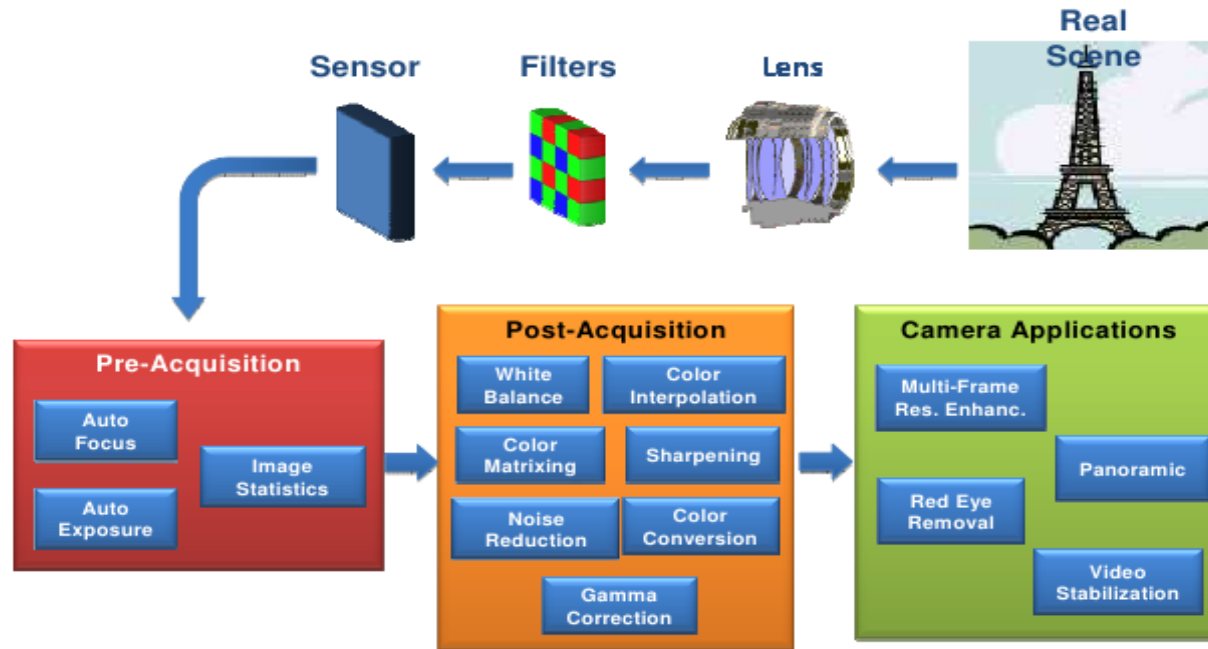


- We need
  - Sensor
  - Transmission
  - Processing
- But this is unfortunately too optimistic!

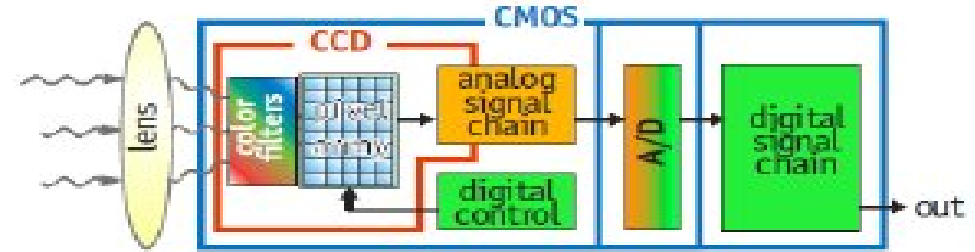
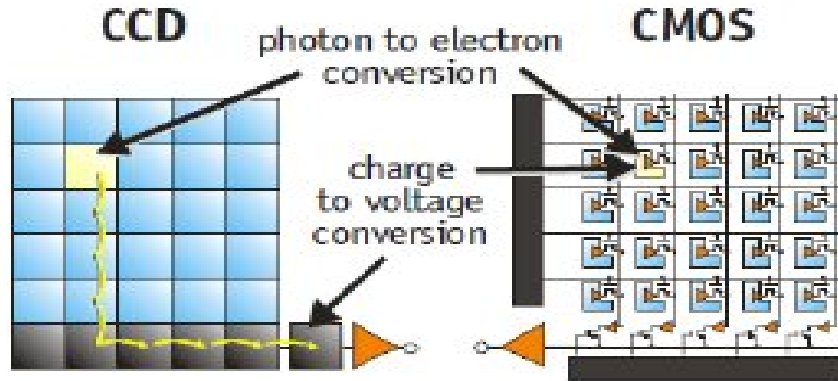




# Actual Acquisition Pipeline



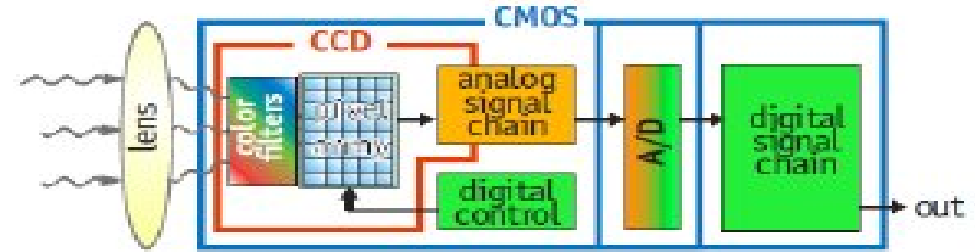
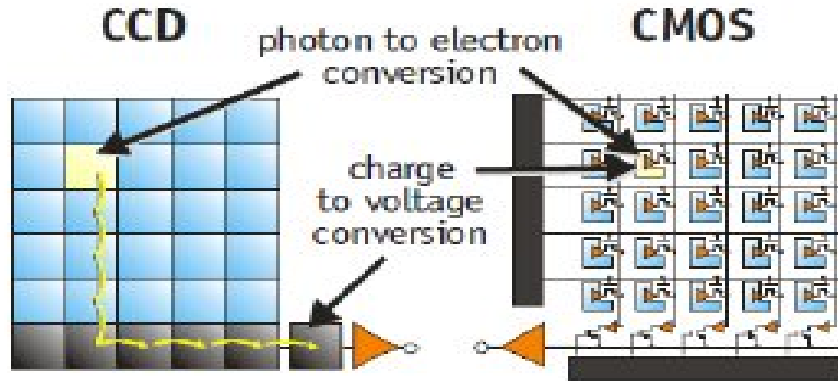
<http://www.dmi.unict.it/~battiato/CVision1011/CVision1011.htm>



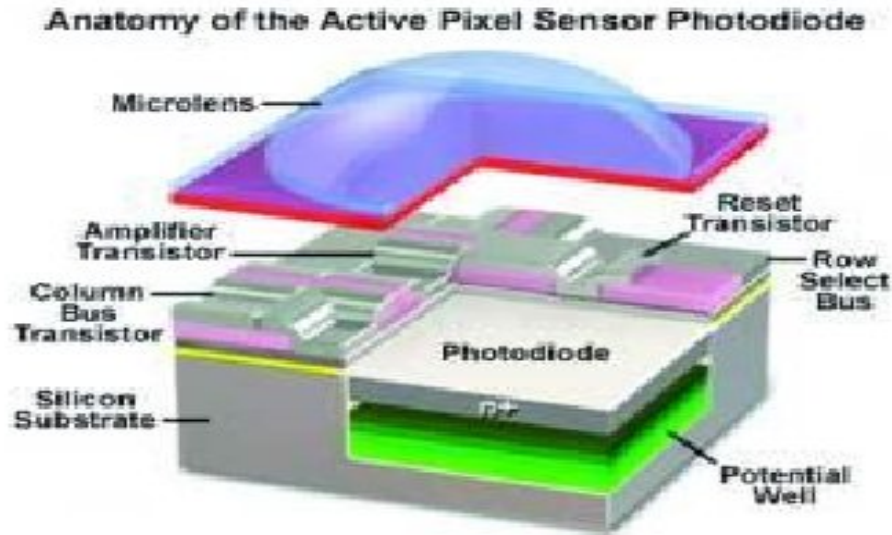
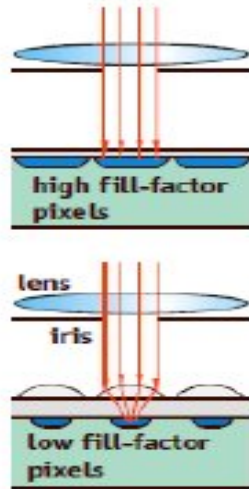
## ■ Charge-Coupled Device:

- Charge is actually transported across the chip and read at one corner of the array and converted (**analog sensor!**)
- Usage of a special manufacturing process to create the ability to transport charge across the chip without distortion.
- Higher Fill Factor

# CCD vs CMOS



- Complimentary Metal-Oxide Semiconductor:
  - Several transistors at each pixel amplify and move the charge using more traditional wires
  - It is more flexible because each pixel can be read individually
  - Usage of the same traditional manufacturing processes to make most microprocessors.
  - Easy integration
  - Lower Fill Factor

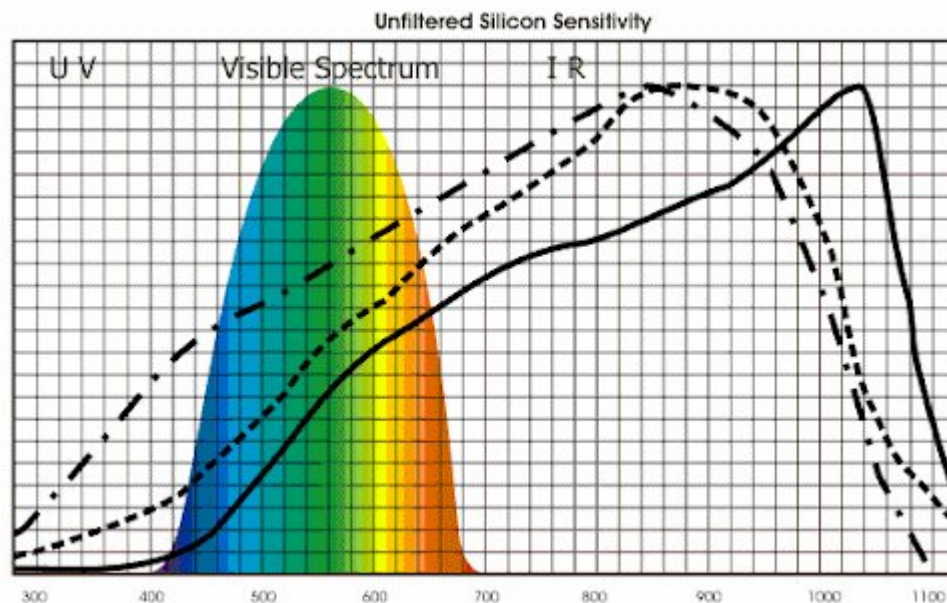
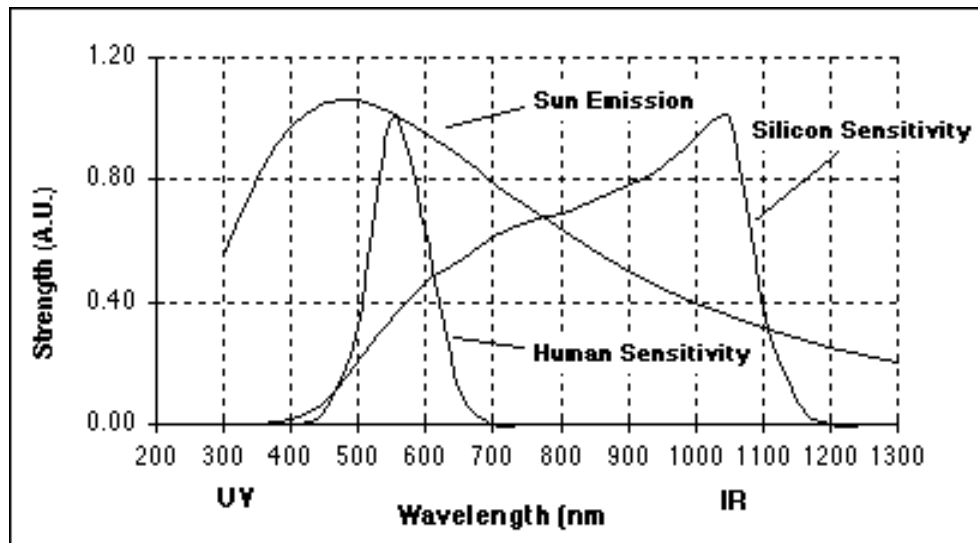


To compensate for lower fill factor (typically 30-50%), most CMOS sensors use microlenses, individual lenses deposited on the surface of each pixel to focus light on the photosensitive area. Microlenses can boost effective fill factor to approximately 70%, improving sensitivity (but not charge capacity) considerably.

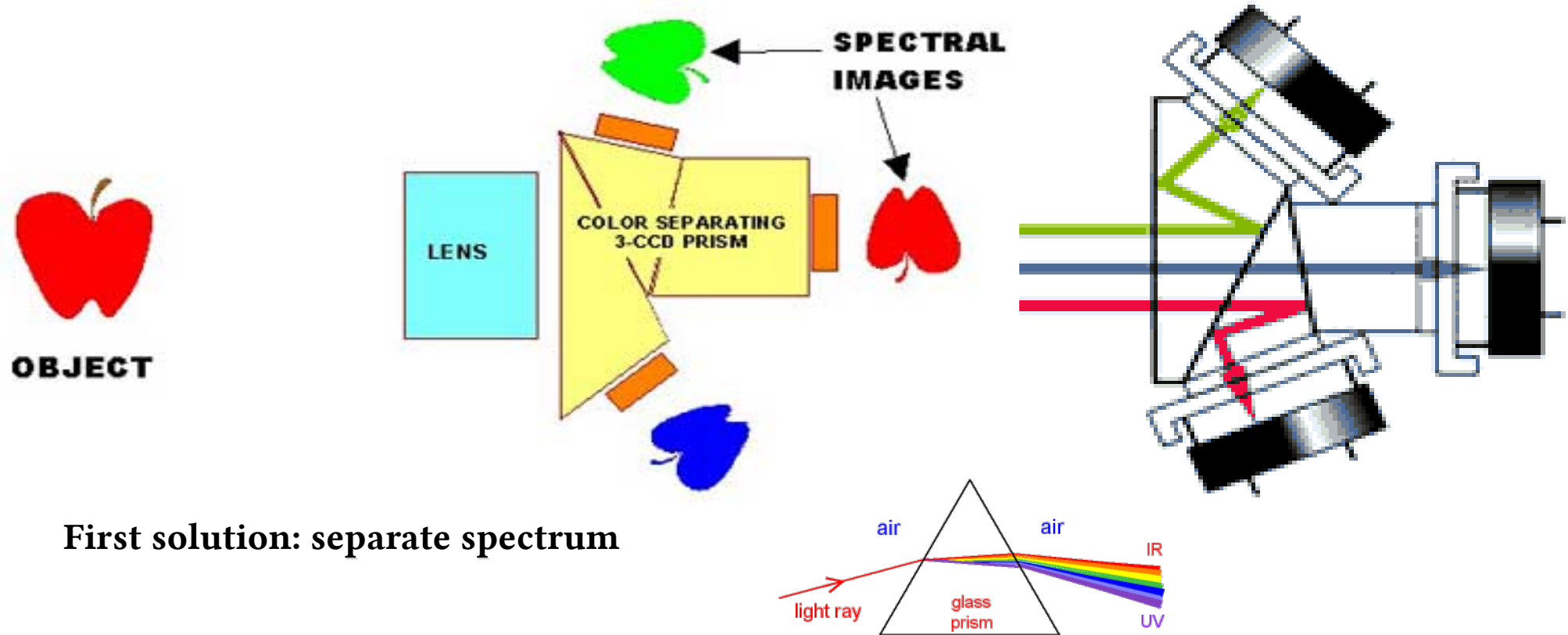


# CMOS sensitivity

- Also valid for CCD



# Color by separating spectrum

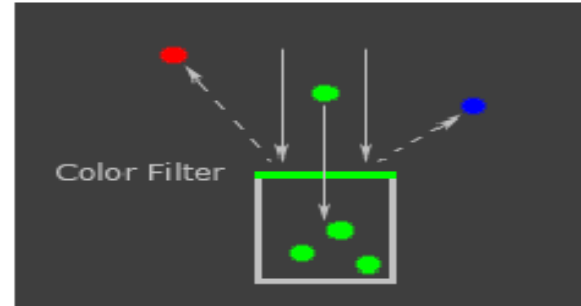
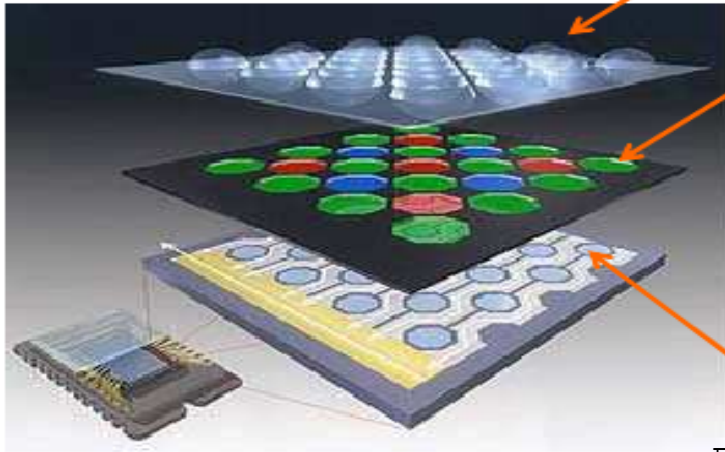


First solution: separate spectrum

# Color Field Array

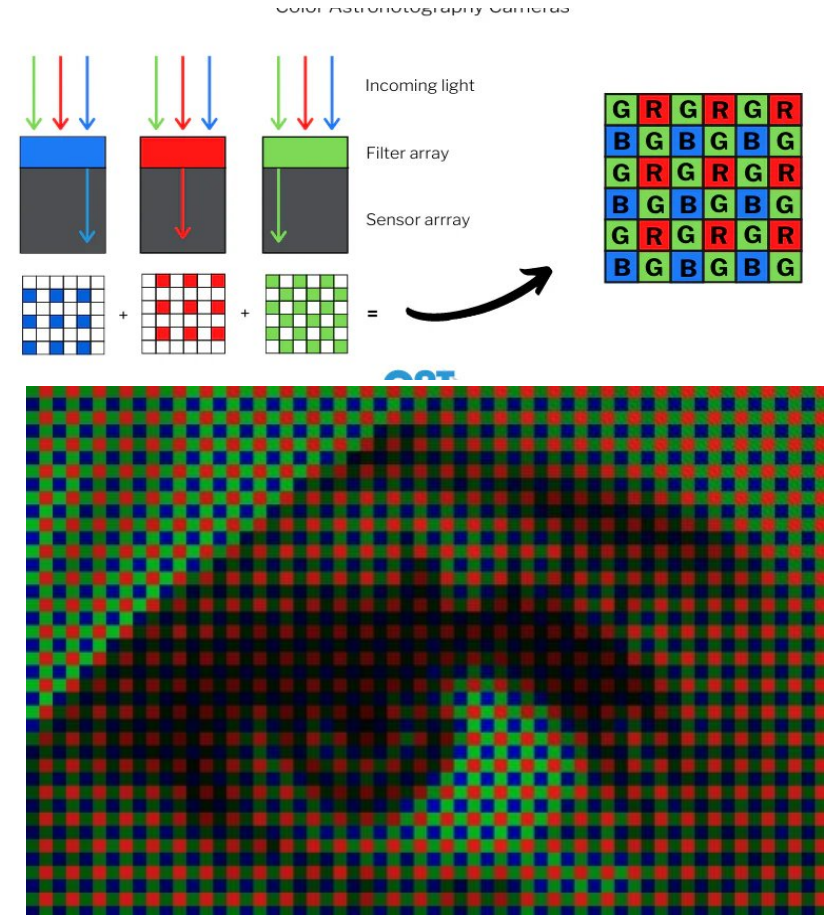
Microlenses to focus spectrum as seen before

Color Field Array is a filter array

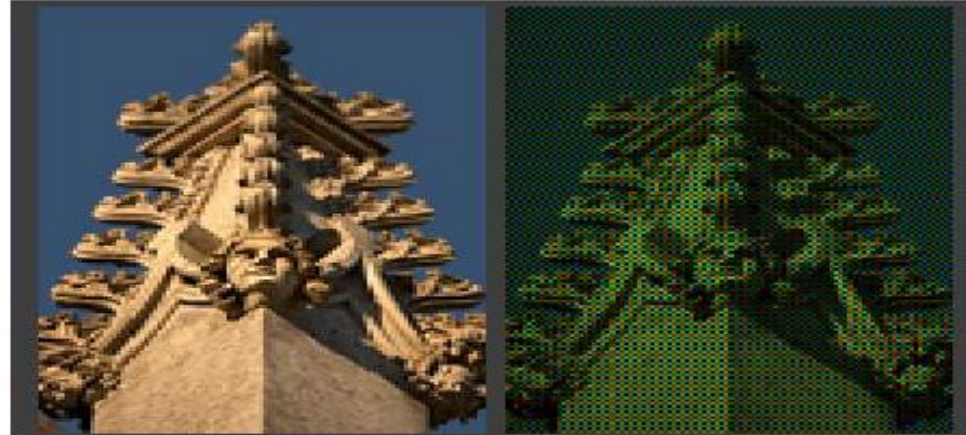
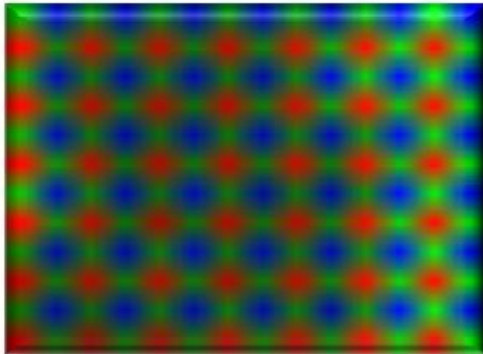
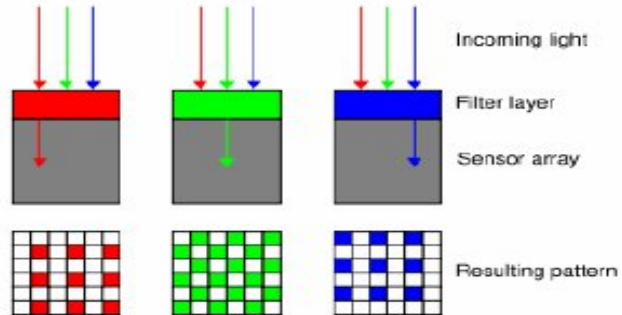


Each photoreceptors get a specific color energy

- Bayer (RGGB, BGGR, GRBG, GBRG)
- RGB+W (RGB + luminance)
- CYGM
  - Also CMY or CMYW
- RCCC (1 red filter, 3 luminance)
- RCCB
- RGB+NIR



# Color Field Array



**Real Scene...**

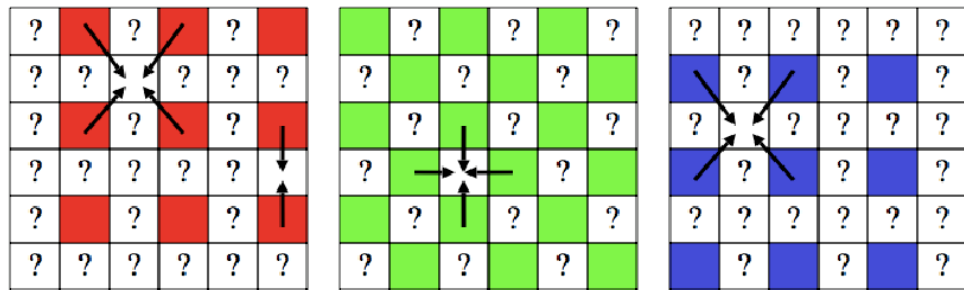
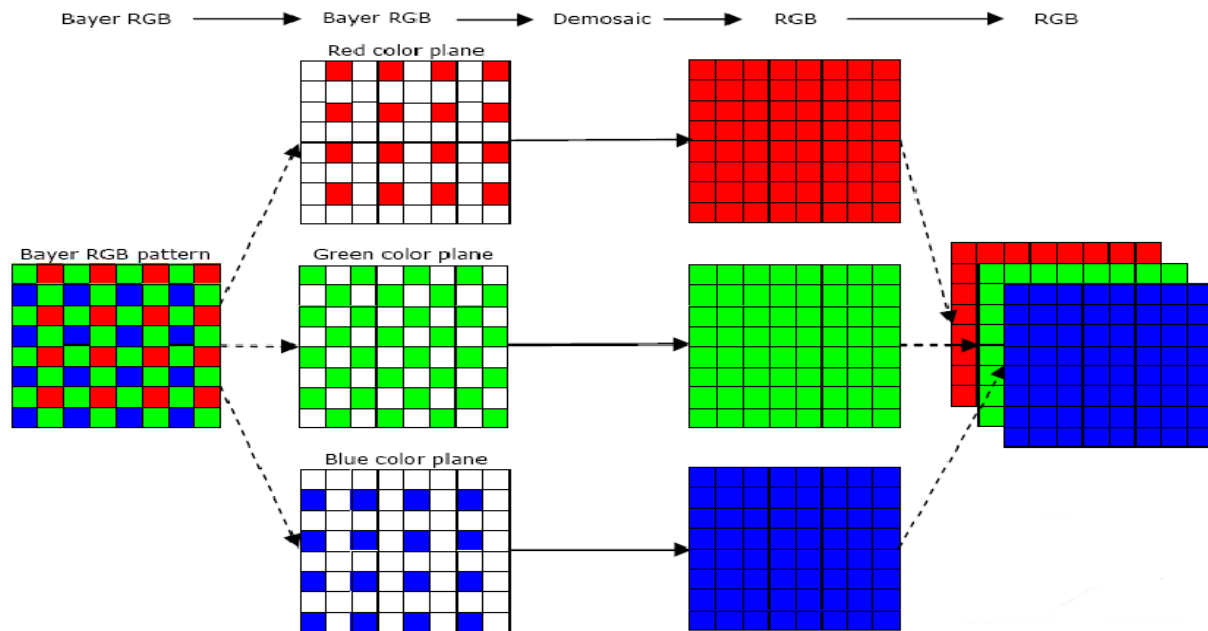
**...as seen by the sensor.**

<http://www.dmi.unict.it/~battiato/CVision1011/CVision1011.htm>

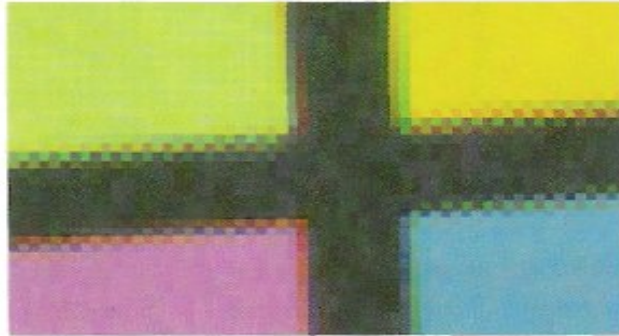


- Simple  $\rightarrow$  interpolation is used for averaging color values
- Downsample  $\rightarrow$  use a  $2 \times 2$  block to obtain a RGB pixel
- Edge Sensing  $\rightarrow$  “color” edges are used to improve average

# Simple Demosaicing



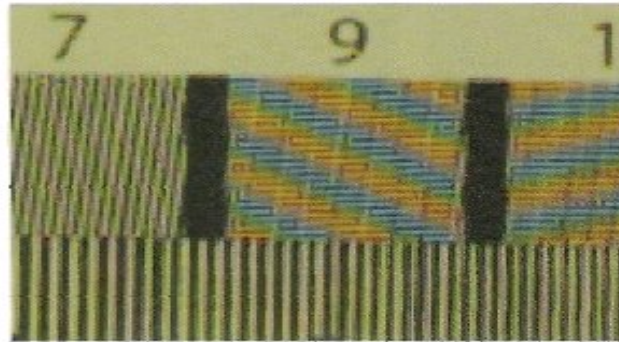
# Demosaicing artifacts



(a)



(b)



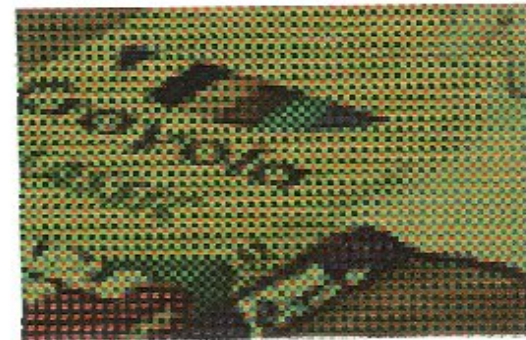
**(a) zipper effects, (b) color shift, (c) aliasing artifacts and (d) blur effects.**

# Demosaicing

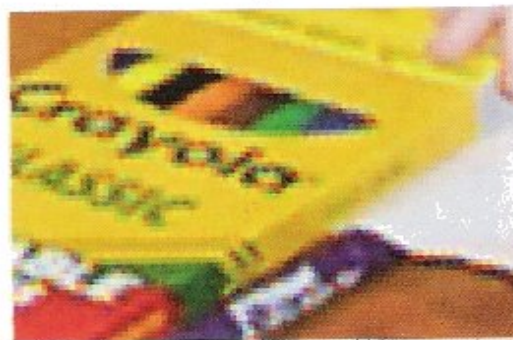
- a) “Grey” levels after color filter array
- b) We can assign a “color” level
- c) Demosaicing
- d) Post Processing



(a)



(b)



(c)



(d)



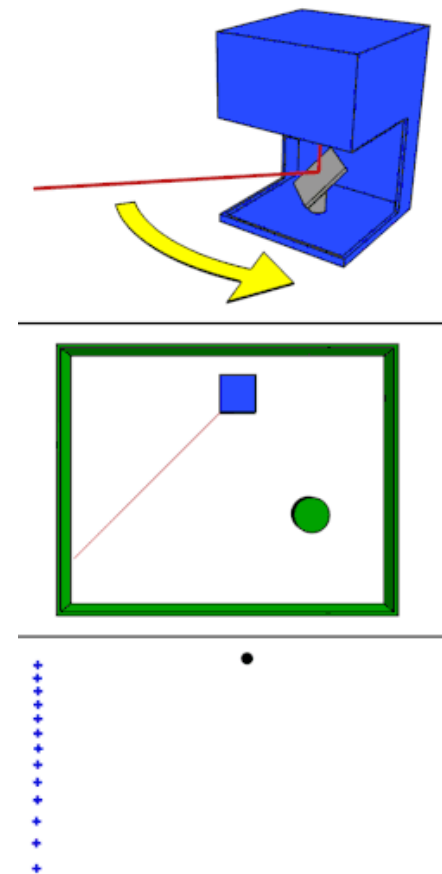
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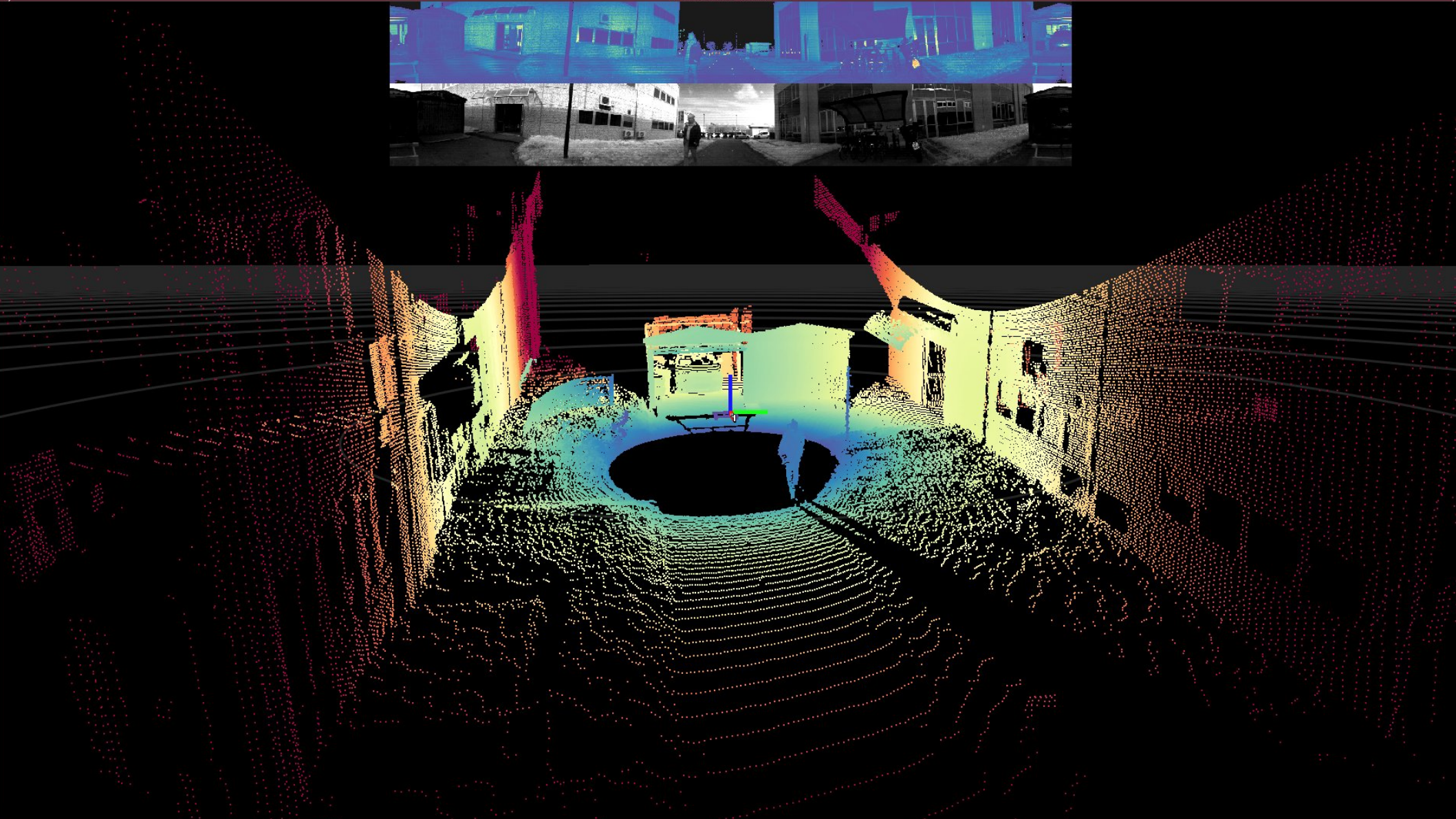
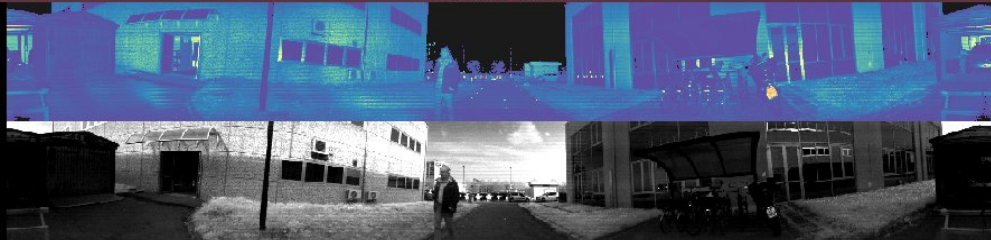
LiDAR



# Light Detection And Ranging

- LiDARs use a pulsed laser light
- They measure echo → direct 3D maps
- Originally mechanical devices, now solid state ones are on the shelves
- 3D maps are not images
  - Actually point clouds
  - Maybe yes!







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# Video & Acquisition Systems